Classical Field Theory — Exercise Sheet 3

Stress, improvements and angular momentum

Solutions

Use $T^\mu{}_\nu = p^\mu \partial_\nu \phi - \delta^\mu_\nu L$, $\eta = \operatorname{diag}(1, -1, \dots, -1)$ and the inverse-metric Hilbert convention of Lecture 3. All quantities below are smooth. The first problem has d spatial coordinates; the remaining problems have three. Problems 1–3 use Lectures 1–3. Problem 4 is a return problem after Lecture 9; its electromagnetic component conventions are supplied in the question.

Exercise 3.1 (Work by an external coupling). On $I \times \mathbb{T}^d$ consider

$$L = \frac{1}{2}\phi_t^2 - \frac{1}{2}|\nabla\phi|^2 - \frac{\lambda(t)}{4}\phi^4,$$

where I is an open interval and λ is prescribed and smooth.

(a) Derive the field equation and energy density. For a classical solution, find the derivative of its total energy, including the term due to explicit time dependence. Decide whether its physical spatial momentum $P^i = \int_{\mathbb{T}^d} T^{0i} \mathrm{d}^d x$ remains constant.

(b) Set $I = (0, \pi/2)$, $\lambda(t) = 1/\sin^2 t$ and $\phi(t, x) = \sin t$. Verify the field equation, compute its energy and check the sign and value of the energy derivative by both methods.

Solution. The momentum is $p^0 = \phi_t$ and $p^i = -\partial_i \phi$. The Euler expression is $E = -\phi_{tt} + \Delta\phi - \lambda\phi^3$, so stationarity gives $\phi_{tt} - \Delta\phi + \lambda\phi^3 = 0$. The energy density and its spatial flux are

$$\rho_{\mathrm{en}} = \frac{1}{2}\phi_t^2 + \frac{1}{2}|\nabla\phi|^2 + \frac{\lambda}{4}\phi^4, \qquad f^i = -\phi_t \partial_i \phi.$$

The explicit derivative of the density is $\partial_t^{\mathrm{explicit}} L = -\lambda'\phi^4/4$. The general identity of Lecture 3 therefore gives, on solutions,

$$\partial_t \rho_{\mathrm{en}} + \partial_i f^i = \frac{\lambda'}{4}\phi^4, \qquad E'(t) = \frac{\lambda'(t)}{4}\int_{\mathbb{T}^d} \phi(t,x)^4 \mathrm{d}^d x.$$

The flux integral vanishes by periodicity. The density has no explicit spatial dependence, so $\partial_\mu T^\mu{}_i = 0$ on solutions. The integrated lower-index momentum is constant, and raising its spatial index changes only its sign. Thus $P^i = \int T^{0i}$ is also constant.

For the proposed homogeneous field, $\phi_{tt} = -\sin t$ and $\lambda\phi^3 = \sin t$. Their sum is zero. Its energy is

$$E(t) = (2\pi)^d \left(\frac{1}{2}\cos^2 t + \frac{1}{4}\sin^2 t\right).$$

Direct differentiation gives $E' = -(2\pi)^d \sin t \cos t/2$. The work formula gives the same value because $\lambda' = -2\cos t/\sin^3 t$ and $\phi^4 = \sin^4 t$. The positive potential does not force conservation when its coefficient is externally changed.

Exercise 3.2 (Angular momentum and weighted decay). For the autonomous scalar density $L = \frac{1}{2}\partial_\mu\phi\partial^\mu\phi - V(\phi)$ on $\mathbb{R}^{1+3}$, raise the second stress index with η. For fixed distinct spatial indices i, j, define

$$M^{\mu ij} = x^i T^{\mu j} - x^j T^{\mu i}.$$

(a) Prove its local conservation on solutions, keeping the derivatives of the explicit coordinate factors.

(b) Prove that $J^{ij}(t) = \int_{\mathbb{R}^3} M^{0ij}\mathrm{d}^3x$ is constant if, on every compact time interval, every component of $T^{\mu\nu}$ is bounded by $C(1+|x|)^{-4-\sigma}$ for some $C, \sigma > 0$. Explain why an unweighted integrability statement for stress alone does not automatically establish the angular charge hypotheses.

Solution. The scalar stress with both indices raised is $T^{\mu\nu} = \partial^\mu\phi\partial^\nu\phi - \eta^{\mu\nu}L$. It is symmetric. Its divergence vanishes on solutions because the density is autonomous. The product rule gives

$$\partial_\mu M^{\mu ij} = T^{ij} - T^{ji} + x^i\partial_\mu T^{\mu j} - x^j\partial_\mu T^{\mu i} = 0.$$

This keeps the coordinate-factor contribution rather than discarding it.

Under the stated bound, $|x^i|$ and $|x^j|$ are at most $1+|x|$, so $|M^{\mu ij}| \le 2C(1+|x|)^{-3-\sigma}$. The spatial integral of this bound is finite: its tail is bounded by a constant times $\int_1^\infty r^{-1-\sigma}\mathrm{d}r$. It is uniform on each compact time interval. Thus the density is integrable on each slice and the spatial flux is spacetime integrable there. Lecture 2's cutoff argument proves that J^{ij} is constant. Mere integrability of T supplies no weighted bound for $|x|T$; the angular current requires that extra control.

Exercise 3.3 (Belinfante improvement from a spin identity). Suppose a smooth tensor $T^{\mu\nu}$ on $\mathbb{R}^{1+3}$ is divergence-free, and a supplied smooth tensor $S^{\lambda\mu\nu}$ satisfies

$$S^{\lambda\mu\nu} = -S^{\lambda\nu\mu}, \qquad \partial_\lambda S^{\lambda\mu\nu} = T^{\nu\mu} - T^{\mu\nu}.$$

This spin identity is a given hypothesis; it is not inferred from scalar Lecture 3. Define

$$B^{\lambda\mu\nu} = \frac{1}{2}\big(S^{\lambda\mu\nu} + S^{\mu\nu\lambda} + S^{\nu\mu\lambda}\big), \qquad \Theta^{\mu\nu} = T^{\mu\nu} + \partial_\lambda B^{\lambda\mu\nu}.$$

Prove that Θ is symmetric and divergence-free. Whenever the slice integrals exist, find the difference between its integrated four-momentum and that of T. Give a sufficient boundary condition for equality, and explain why local improvement alone is insufficient.

Solution. Using antisymmetry in the last two slots of S gives

$$B^{\mu\lambda\nu} = \frac{1}{2}\big(-S^{\mu\nu\lambda} - S^{\lambda\mu\nu} - S^{\nu\mu\lambda}\big) = -B^{\lambda\mu\nu}.$$

Subtracting the two expressions with μ and ν interchanged gives $B^{\lambda\mu\nu} - B^{\lambda\nu\mu} = S^{\lambda\mu\nu}$. Consequently the supplied spin identity implies

$$\Theta^{\mu\nu} - \Theta^{\nu\mu} = T^{\mu\nu} - T^{\nu\mu} + \partial_\lambda S^{\lambda\mu\nu} = 0.$$

The added divergence has identically zero divergence because its first two slots are antisymmetric and mixed derivatives commute. Hence $\partial_\mu\Theta^{\mu\nu} = \partial_\mu T^{\mu\nu} = 0$.

On a spatial cube Q_R, coordinate FTC gives the momentum difference

$$\int_{Q_R} (\Theta^{0\nu} - T^{0\nu})\mathrm{d}^3x = \int_{\partial Q_R} B^{i0\nu} n_i \,\mathrm{d}S.$$

There is no $\partial_t B^{00\nu}$ term, since $B^{00\nu} = 0$. If $B^{i0\nu}$ is compactly supported in space the boundary integral vanishes for large R. Periodicity also makes opposite faces cancel. Under either condition, provided the original charges exist, the improved and original momenta agree. Without control of this boundary term, the local identities do not imply equality of integrated charges.

Exercise 3.4 (Maxwell Hilbert stress, after Lecture 9). Return to this problem after Lecture 9. Let $F_{\mu\nu} = -F_{\nu\mu}$ be held fixed with both indices down while a Lorentz metric is varied. Use the density and Hilbert convention

$$L = -\frac{1}{4} g^{\mu\rho} g^{\nu\sigma} F_{\mu\nu} F_{\rho\sigma}, \qquad \delta S = \frac{1}{2} \int T_{\mu\nu} k^{\mu\nu} \mathrm{vol}_g, \quad k^{\mu\nu} = \delta g^{\mu\nu}.$$

The volume variation is $\delta \mathrm{vol}_g = -\frac{1}{2} g_{\mu\nu} k^{\mu\nu} \mathrm{vol}_g$. Derive $T_{\mu\nu}$ by varying both inverse metrics. At $g = \eta$, set $F_{i0} = E_i$ and $F_{ij} = \epsilon_{ijk} B_k$, with $\epsilon_{123} = 1$. Compute T_{00} and T^{i0} in terms of the vectors $\mathbf{E}$ and $\mathbf{B}$. Evaluate them for $\mathbf{E} = (1, 2, 0)$ and $\mathbf{B} = (0, 0, 3)$.

Solution. Varying each inverse metric in the stated contraction and relabelling the summed indices gives two equal terms. Thus

$$\delta L = -\frac{1}{2} F_{\mu\alpha} F_{\nu}{}^{\alpha} k^{\mu\nu}.$$

Multiplying by the varied volume gives

$$\delta(L \mathrm{vol}_g) = \frac{1}{2} \left[-F_{\mu\alpha} F_{\nu}{}^{\alpha} + \frac{1}{4} g_{\mu\nu} F_{\alpha\beta} F^{\alpha\beta} \right] k^{\mu\nu} \mathrm{vol}_g.$$

Comparison with the plus Hilbert convention identifies the tensor inside the brackets as $T_{\mu\nu}$.

The supplied electromagnetic convention gives $F_{\alpha\beta} F^{\alpha\beta} = 2(|\mathbf{B}|^2 - |\mathbf{E}|^2)$ and $F_{0\alpha} F_0{}^{\alpha} = -|\mathbf{E}|^2$. Therefore

$$T_{00} = \frac{1}{2} (|\mathbf{E}|^2 + |\mathbf{B}|^2).$$

For a spatial index the trace term is zero, and direct contraction gives $T_{i0} = -\epsilon_{ijk} E_j B_k$. Raising that spatial index gives $T^{i0} = \epsilon_{ijk} E_j B_k = (\mathbf{E} \times \mathbf{B})_i$. In the example the squared magnitudes are 5 and 9, so $T_{00} = 7$. The cross product is $(2 \cdot 3, -1 \cdot 3, 0) = (6, -3, 0)$. These signs follow from the inverse-metric convention, not a choice to make energy positive after the calculation.